Wireshark Traffic Analysis Report

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Project: Network Traffic Analysis Using Wireshark

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1. Objective

The objective of this project was to capture and analyze live network traffic using Wireshark in order to understand how common network protocols operate and interact in real-world environments.

2. Tools Used

- Wireshark
- macOS Terminal

3. Methodology

Network traffic was captured using Wireshark on the primary wireless interface. Traffic was generated by browsing websites and executing network commands such as `ping` and `nslookup`.

Filters were applied within Wireshark to isolate specific protocols including DNS, TCP, TLS, and ICMP.

4. Analysis

4.1 DNS Analysis

DNS traffic was analyzed to observe how domain names are resolved into IP addresses. Queries were captured for domains such as `grammarly.com`, `cursor.sh`, and `xfinity.com`.

Both A (IPv4) and AAAA (IPv6) records were observed, along with multiple IP responses, indicating load balancing. Additionally, DNS over HTTPS (DoH) traffic was identified, demonstrating modern secure DNS resolution techniques.

4.2 TCP Handshake Analysis

TCP packets containing SYN and SYN-ACK flags were captured, representing the initial stages of the TCP three-way handshake.

This process ensures reliable communication between client and server before data transmission. The traffic was observed over port 443, indicating HTTPS connections.

4.3 TLS / HTTPS Analysis

TLS traffic was analyzed to understand encrypted communication. The capture included key stages such as Client Hello, Server Hello, certificate exchange, and key exchange.

Following the handshake, packets were labeled as “Application Data,” indicating encrypted traffic. TLSv1.3 was observed, highlighting the use of modern encryption standards.

4.4 ICMP Analysis

ICMP echo request and reply packets were captured using the ping command. These packets confirmed successful communication between the local machine and a remote server.

The TTL values observed indicate the number of hops taken across the network.

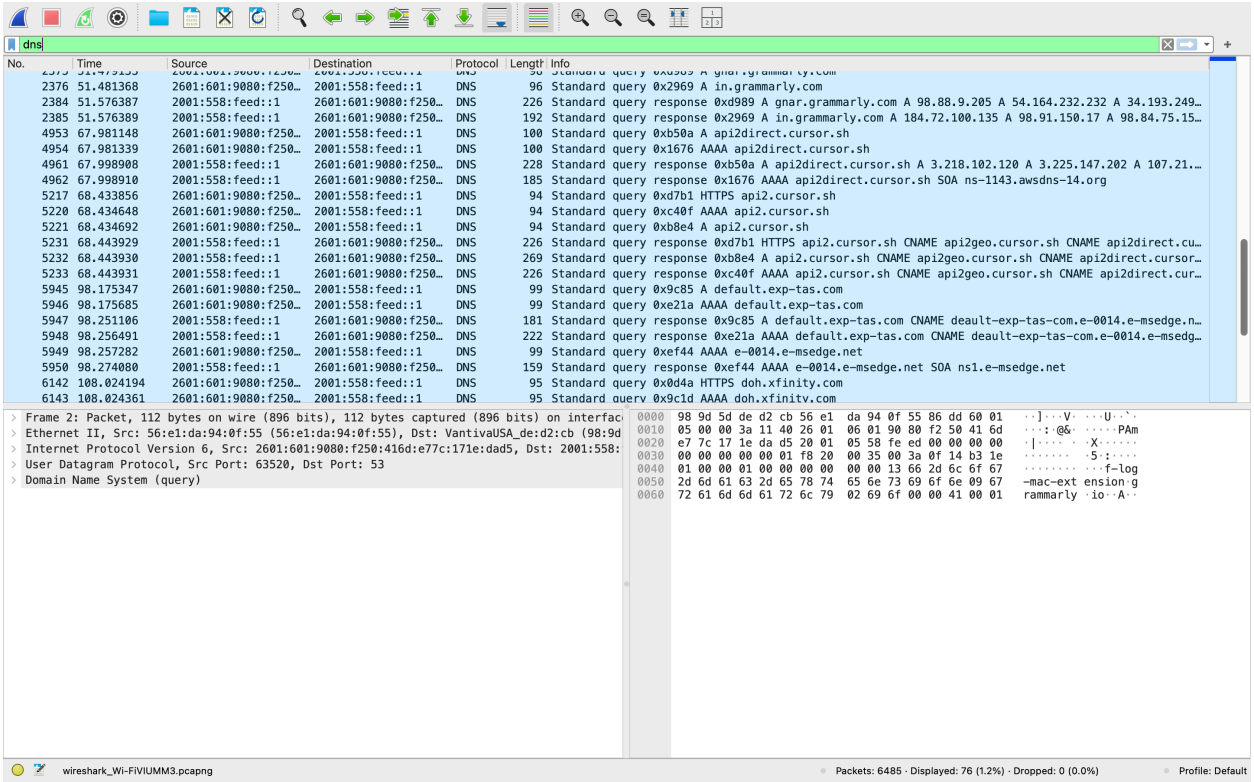
5. Key Findings

- DNS is used to resolve domain names into IP addresses
- TCP ensures reliable communication through a three-way handshake
- TLS encrypts network traffic, preventing visibility into transmitted data
- ICMP is used for network diagnostics and connectivity testing

6. Conclusion

This project provided hands-on experience with network traffic analysis and improved understanding of how core protocols operate in real-world environments. The ability to capture and interpret network data is a critical skill in cybersecurity, particularly for roles such as SOC analyst or network security engineer.

7. Appendix



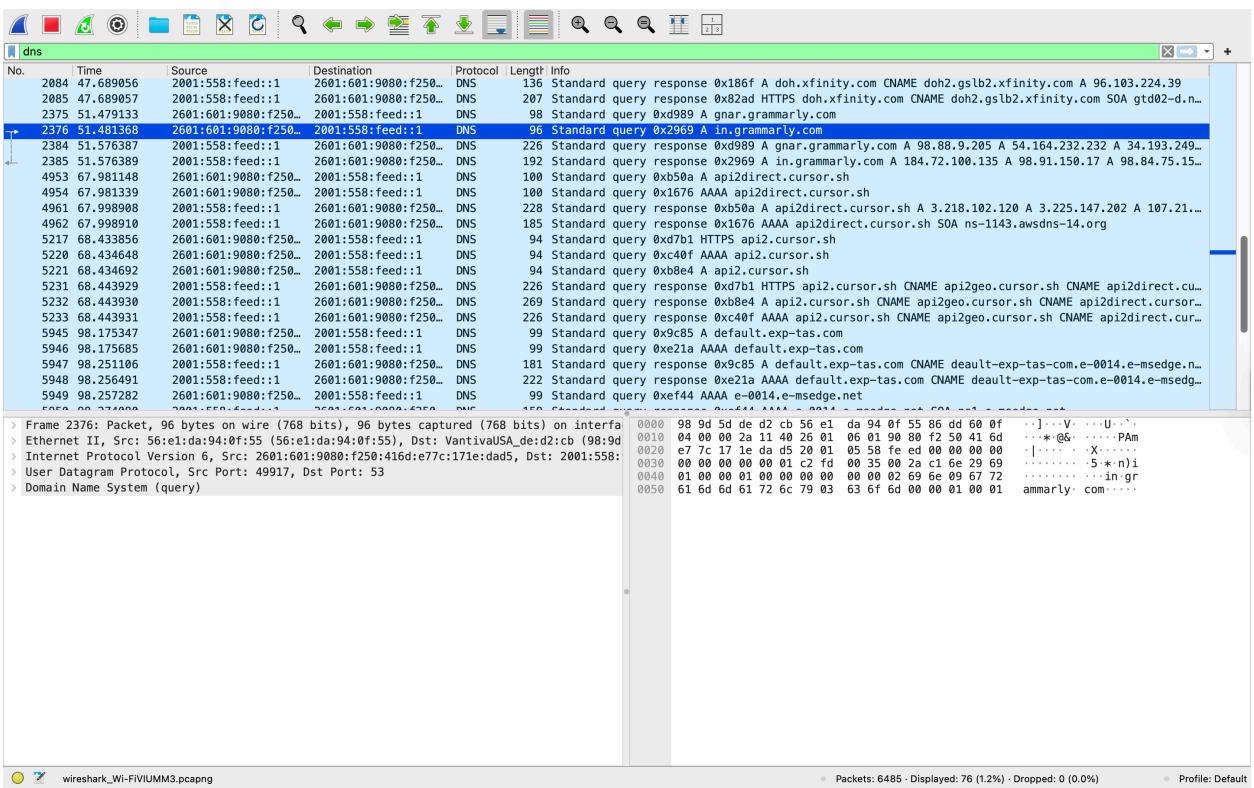


Figure 1: DNS Analysis

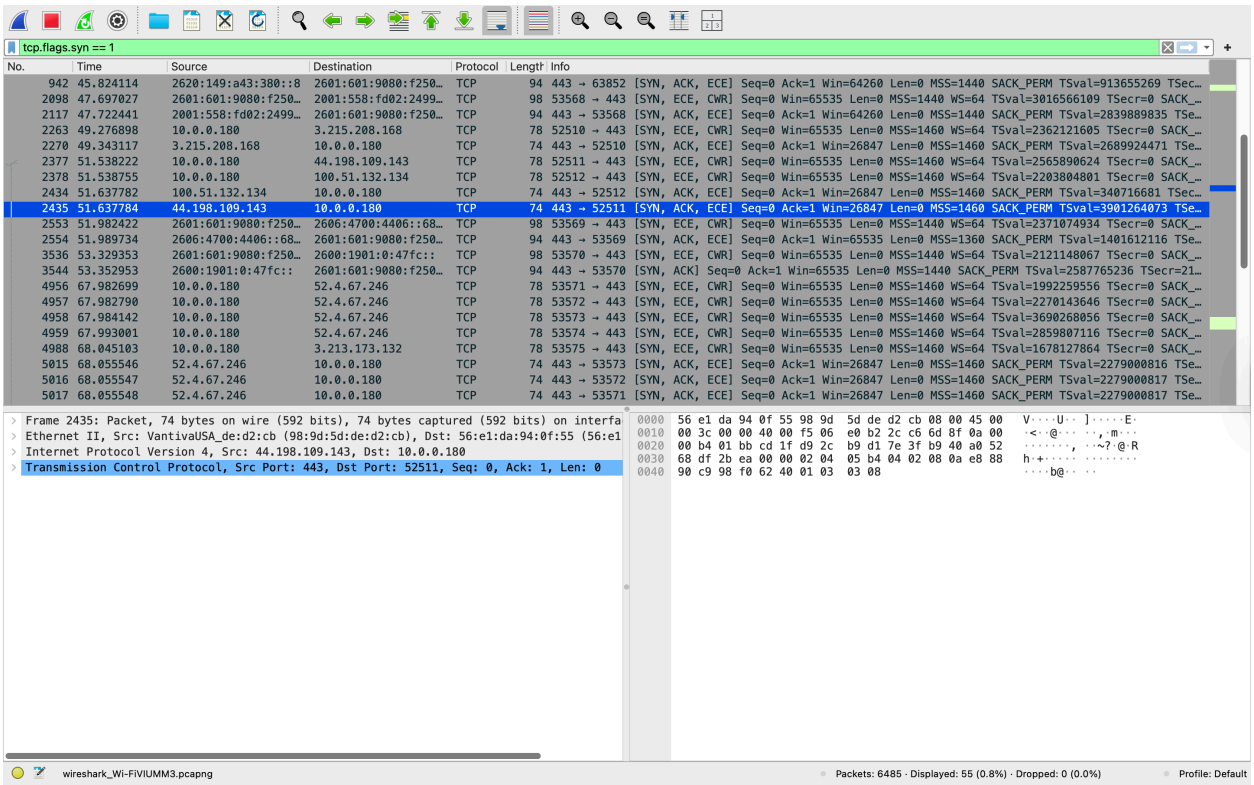


Figure 2: TCP Handshake analysis

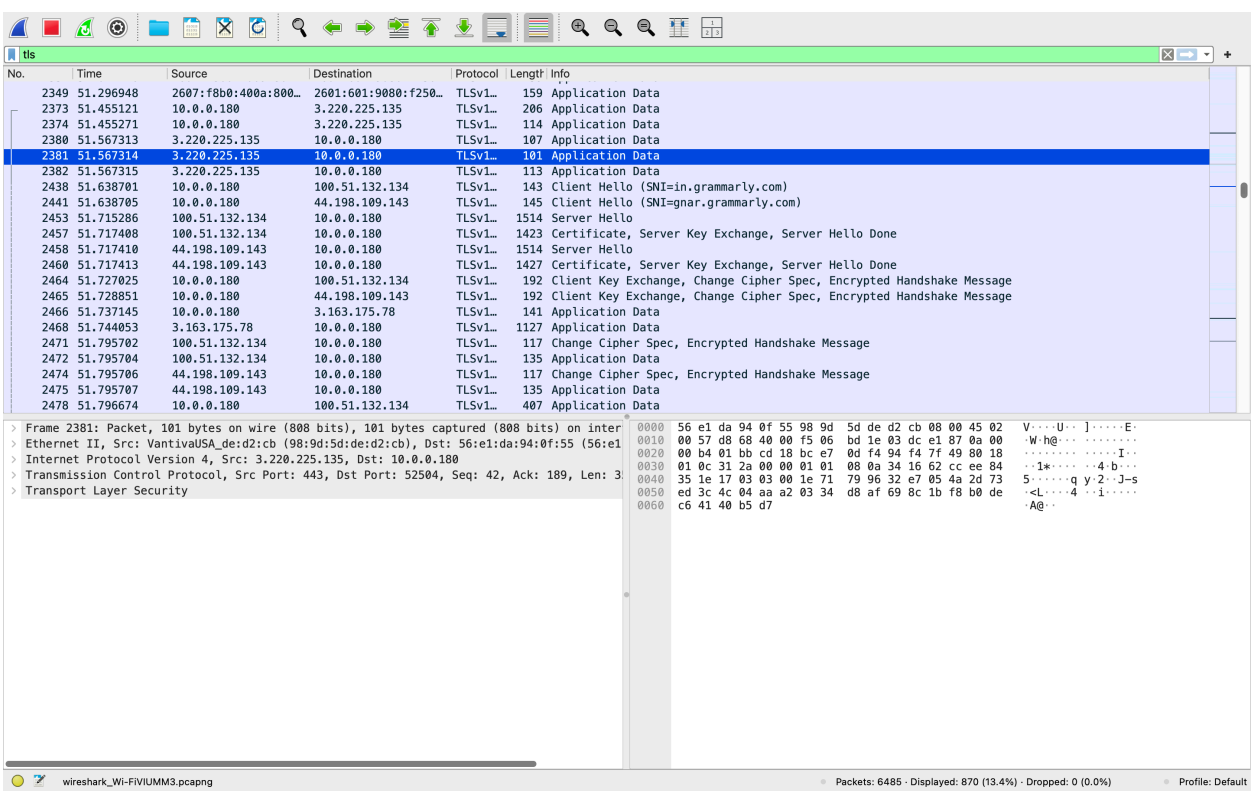
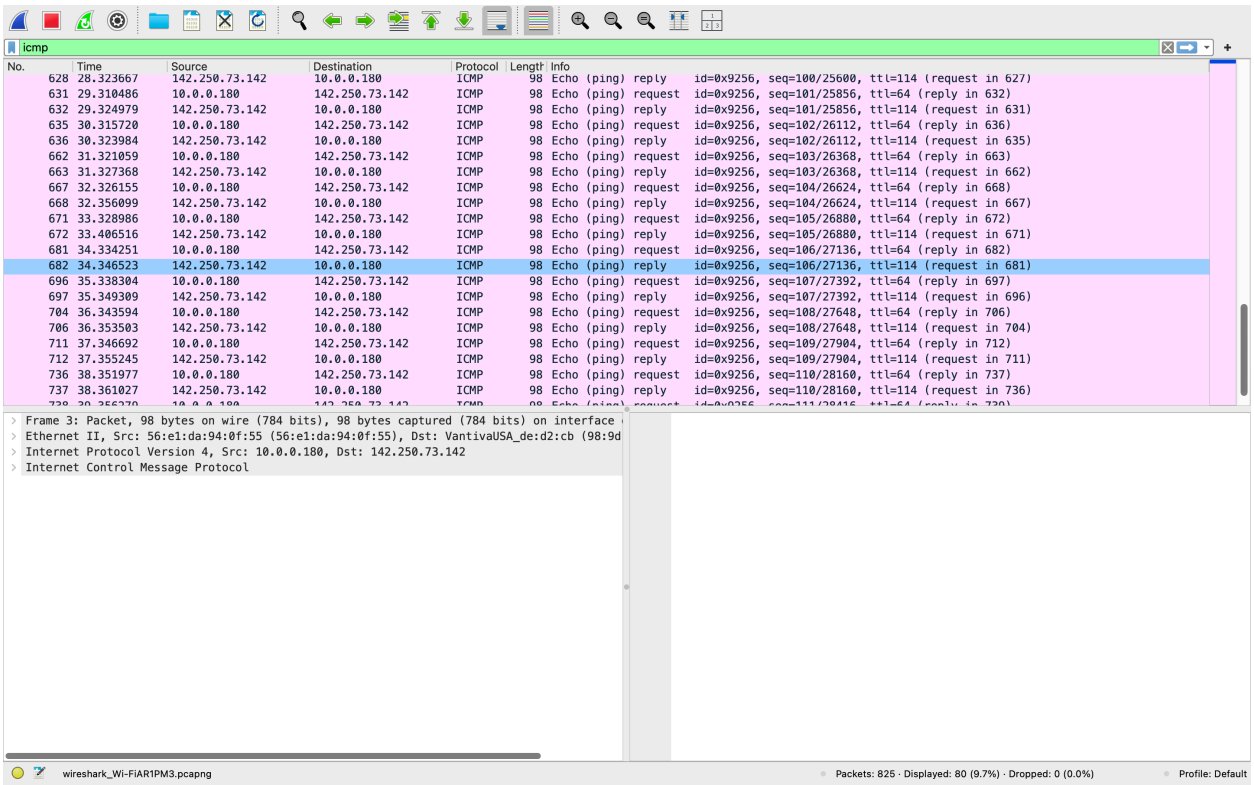


Figure 3: TLS traffic analysis



icmp

No.	Time	Source	Destination	Protocol	Length	Info
628	28.323667	142.250.73.142	10.0.0.180	ICMP	98	Echo (ping) reply id=0x9256, seq=100/25600, ttl=114 (request in 627)
631	29.310486	10.0.0.180	142.250.73.142	ICMP	98	Echo (ping) request id=0x9256, seq=101/25856, ttl=64 (reply in 632)
632	29.324979	142.250.73.142	10.0.0.180	ICMP	98	Echo (ping) reply id=0x9256, seq=101/25856, ttl=114 (request in 631)
635	30.315720	10.0.0.180	142.250.73.142	ICMP	98	Echo (ping) request id=0x9256, seq=102/26112, ttl=64 (reply in 636)
636	30.323984	142.250.73.142	10.0.0.180	ICMP	98	Echo (ping) reply id=0x9256, seq=102/26112, ttl=114 (request in 635)
662	31.321059	10.0.0.180	142.250.73.142	ICMP	98	Echo (ping) request id=0x9256, seq=103/26368, ttl=64 (reply in 663)
663	31.327368	142.250.73.142	10.0.0.180	ICMP	98	Echo (ping) reply id=0x9256, seq=103/26368, ttl=114 (request in 662)
667	32.326155	10.0.0.180	142.250.73.142	ICMP	98	Echo (ping) request id=0x9256, seq=104/26624, ttl=64 (reply in 668)
668	32.356099	142.250.73.142	10.0.0.180	ICMP	98	Echo (ping) reply id=0x9256, seq=104/26624, ttl=114 (request in 667)
671	33.328986	10.0.0.180	142.250.73.142	ICMP	98	Echo (ping) request id=0x9256, seq=105/26880, ttl=64 (reply in 672)
672	33.406516	142.250.73.142	10.0.0.180	ICMP	98	Echo (ping) reply id=0x9256, seq=105/26880, ttl=114 (request in 671)
681	34.334251	10.0.0.180	142.250.73.142	ICMP	98	Echo (ping) request id=0x9256, seq=106/27136, ttl=64 (reply in 682)
682	34.346523	142.250.73.142	10.0.0.180	ICMP	98	Echo (ping) reply id=0x9256, seq=106/27136, ttl=114 (request in 681)
696	35.338304	10.0.0.180	142.250.73.142	ICMP	98	Echo (ping) request id=0x9256, seq=107/27392, ttl=64 (reply in 697)
697	35.349309	142.250.73.142	10.0.0.180	ICMP	98	Echo (ping) reply id=0x9256, seq=107/27392, ttl=114 (request in 696)
704	36.343594	10.0.0.180	142.250.73.142	ICMP	98	Echo (ping) request id=0x9256, seq=108/27648, ttl=64 (reply in 706)
706	36.353503	142.250.73.142	10.0.0.180	ICMP	98	Echo (ping) reply id=0x9256, seq=108/27648, ttl=114 (request in 704)
711	37.346692	10.0.0.180	142.250.73.142	ICMP	98	Echo (ping) request id=0x9256, seq=109/27904, ttl=64 (reply in 712)
712	37.355245	142.250.73.142	10.0.0.180	ICMP	98	Echo (ping) reply id=0x9256, seq=109/27904, ttl=114 (request in 711)
736	38.351977	10.0.0.180	142.250.73.142	ICMP	98	Echo (ping) request id=0x9256, seq=110/28160, ttl=64 (reply in 737)
737	38.361027	142.250.73.142	10.0.0.180	ICMP	98	Echo (ping) reply id=0x9256, seq=110/28160, ttl=114 (request in 736)

```
> Frame 3: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface
> Ethernet II, Src: 56:e1:da:94:0f:55 (56:e1:da:94:0f:55), Dst: VantivaUSA_de:d2:cb (98:9d
> Internet Protocol Version 4, Src: 10.0.0.180, Dst: 142.250.73.142
> Internet Control Message Protocol
```

wireshark_Wi-FiAR1PM3.pcapng · Packets: 825 · Displayed: 80 (9.7%) · Dropped: 0 (0.0%) · Profile: Default

Figure 4: ICMP analysis